



Cafe Shop Management System

Software Requirement Specification

Project Code: CSMS

Document Code: CSMS-SRS – v1.0

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I. Record of Changes

Date	A* M, D	In charge	Change Description
12/06/2026	A	Project Team	Initial version of the SRS for the Cafe Shop Management System.

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Product Overview

The Cafe Shop Management System (CSMS) is a software application designed to support the daily operations of a coffee shop, including menu management, order taking, billing and payment, inventory tracking, table management, and sales reporting. It replaces manual, paper-based processes with a centralized digital platform that improves order accuracy, speeds up service, and gives the owner clear visibility into sales and stock.

The system is used by cafe sales agent at the point of sale and by management for back-office tasks. The context diagram below illustrates the boundary of the system, the external entities (actors and external systems) that interact with it, and the main information flows between them.

Figure 1-1 below presents the context diagram for release 1.0 of the Cafe Shop Management System.

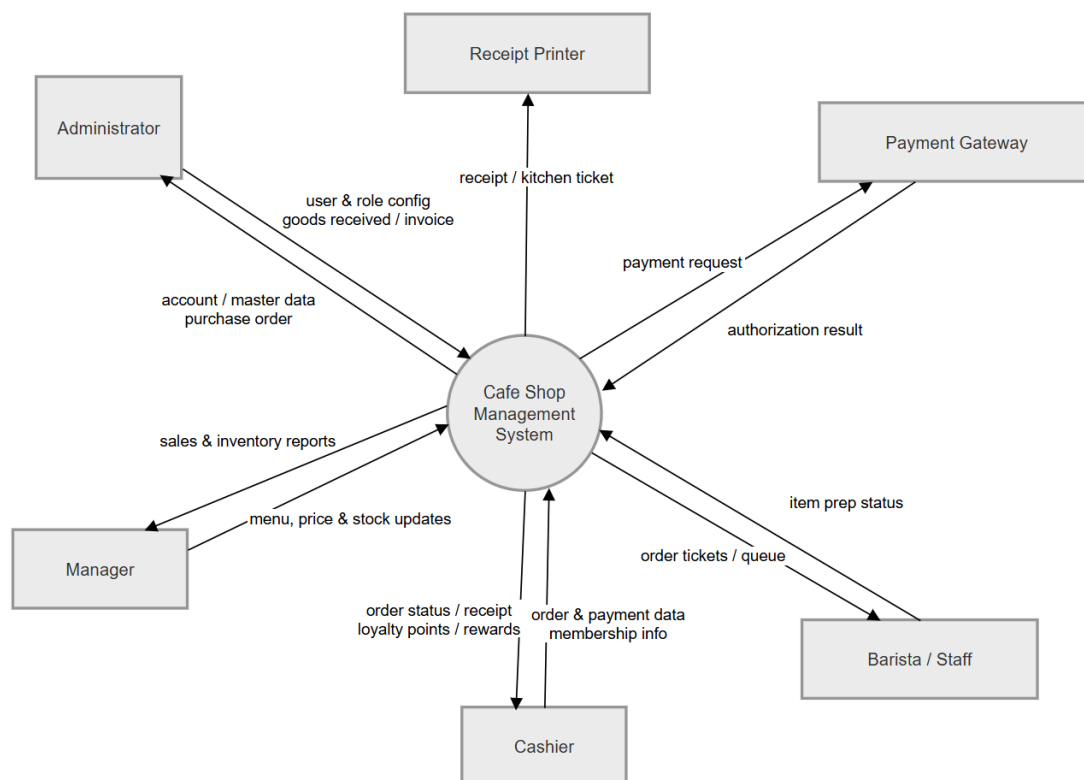


Figure 1-1: Cafe Shop Management System – Context Diagram

As shown in Figure 1-1, the system interfaces with the following external entities:

- **Payment Gateway:** an external service that authorizes and processes card and e-wallet payments submitted by the cashier.

- **Receipt Printer:** a hardware device that prints customer receipts and kitchen order tickets generated by the system.

2. User Requirements

2.1 Actors

#	Actor	Description
1	Administrator	Configures the system, manages user accounts and roles, and sets up master data such as products, categories, and tables.
2	Manager	Oversees daily operations, manages the menu and pricing, monitors inventory, and views sales and performance reports.
3	Cashier	Creates and edits customer orders, processes payments, applies discounts, and prints receipts at the point of sale.
4	Barista/Staff	Receives order tickets, prepares drinks and food, and updates the preparation status of order items.
5	Payment Gateway	External system that authorizes and settles electronic (card / e-wallet) payments.
6	Receipt Printer	External hardware device that prints customer receipts and kitchen order tickets generated by the system.

2.2 Use Cases

2.2.1 Diagram(s)

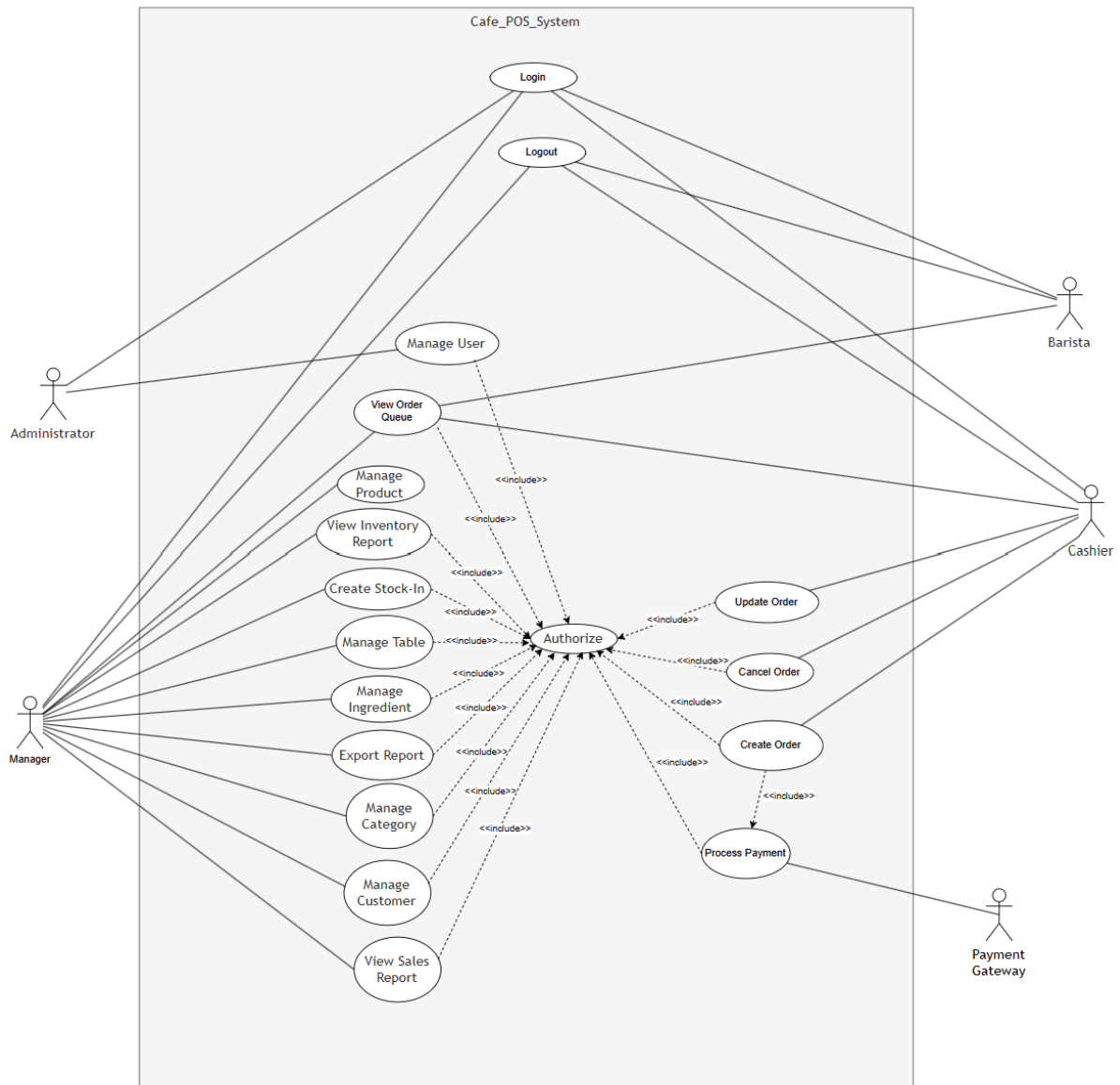


Figure 2-1: Use case diagram of the Cafe Shop Management System

2.2.2 Descriptions

ID	Group function	Use Case	Actors	Use Case Description & Main Flow
01	Authentication	Login	Administrator, Manager,	Authenticates the user with username and password to access permitted functions.

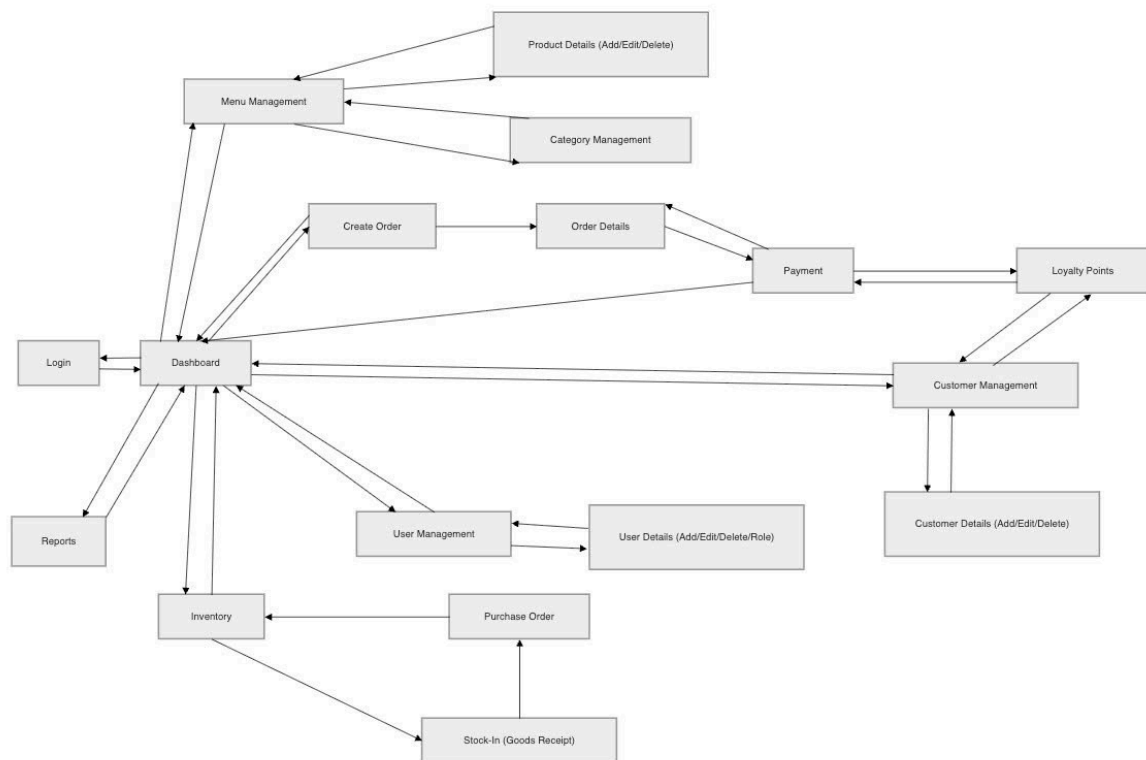
			Cashier, Barista	
02	Authentication	Logout	Administrator, Manager, Cashier, Barista	Terminates the current session and returns the user to the login screen.
03	Authorization	Authorize	Administrator, Manager, Cashier, Barista	Verifies user permissions for specific features based on assigned roles.
04	Administration	Manage User	Administrator	Creates, edits, and removes user accounts; assigns roles.
05	Order Management	Create Order	Cashier	Opens a new order, adds items with options, and assigns to a table or takeaway.
06	Order Management	Update Order	Cashier	Adds, removes, or modifies items on an open order before payment.
07	Order Management	Cancel Order	Cashier	Cancels an open order and logs the reason.
08	Payment	Process Payment	Cashier, Payment Gateway	Settles the bill, applies discounts, processes loyalty points, and prints the receipt.
09	Order Queue	View Order Queue	Barista, Cashier, Manager	Displays open orders and tracks item preparation status.
10	Menu Management	Manage Product	Manager	Manages product catalog (add/edit/delete/view).
11	Menu Management	Manage Category	Manager	Manages product categories (add/edit/delete).
12	Inventory	Manage Ingredient	Manager	Manages ingredient stock levels and thresholds.
13	Inventory	Create Stock-In	Manager	Records received goods against a purchase order.
14	Inventory	View Inventory Report	Manager	Reviews stock levels and identifies low-stock items.

15	Table Management	Manage Table	Manager	Configures cafe tables (add/edit/delete).
16	Customer	Manage Customer	Manager	Manages customer records and loyalty information.
17	Reporting	View Sales Report	Manager	Generates sales and revenue reports based on date ranges.
18	Reporting	Export Report	Manager	Exports sales reports to external files.

3. Software Features

3.1 Functional Overview

3.1.1 Screens Flow



3.1.2 Screen Descriptions

#	Feature	Screen	Description
1	Authentication	Login	Allows a user to sign in with username and password.

2	Common	Dashboard	Home screen after login; shortcuts to role-permitted features and summary widgets (today's revenue, open orders, low-stock alerts).
3	Order Management	Create Order	Start a new order, add menu items with quantities/options, assign to a table or takeaway.
4	Order Management	Order Details	Shows items, quantities and total; allows editing items before payment.
5	Order Management	View Order Queue	Lists open orders and preparation status; lets baristas mark items in-progress/completed.
6	Payment	Payment	Process the bill, select payment method, apply discount, earn/redeem customer loyalty points, and print receipt.
7	Menu Management	Menu Management	Entry point to manage products and categories.
8	Menu Management	Product Details (Add/Edit/Delete)	Manages individual products.
9	Menu Management	Category Management	Manages categories.
10	Inventory	Inventory	Views stock levels and low-stock alerts.
11	Inventory	Purchase Order	Creates/views purchase orders to suppliers.
12	Inventory	Stock-In (Goods Receipt)	Records received goods against a purchase order.
13	Reporting	Reports	Sales and revenue reports filtered by date range.
14	Administration	User Management	Manages user accounts.
15	Administration	User Details (Add/Edit/Delete/Role)	Manages individual users and roles.
16	Customer	Customer Management	Entry point to manage customers and loyalty.
17	Customer	Customer Details (Add/Edit/Delete)	Manages individual customers.

3.1.3 Screen Authorization

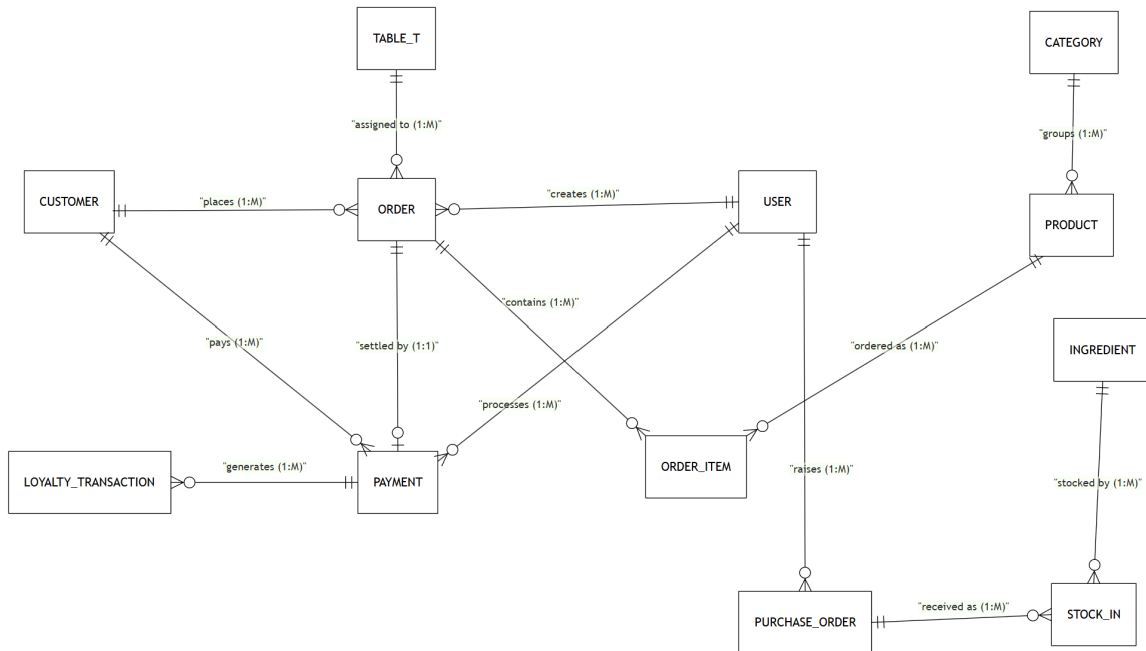
#	Screen	Administrator	Manager	Cashier	Barista
1	Login / Logout	X	X	X	X
2	Dashboard	X	X	X	X
3	Create Order			X	

4	Order Details			X	
5	View Order Queue		X	X	X
6	Payment			X	
7	Menu Management		X		
7.1 Product Details (Add/Edit/Delete)			X		
7.2 Category Management			X		
8	Inventory		X		
8.1 Purchase Order			X		
8.2 Stock-In (Goods Receipt)			X		
9	Reports		X		
10	User Management	X			
10.1 User Details (Add/Edit/Delete/Role)		X			
11	Customer Management		X		
11.1 Customer Details (Add/Edit/Delete)			X		

3.1.4 Non-Screen Functions

#	Feature	System Function	Description
1	Reporting	Daily Sales Aggregation	Scheduled end-of-day job aggregating sales into reporting tables.
2	Inventory	Low-Stock Alert	Background service notifying the manager when an ingredient falls below its reorder threshold.
3	Inventory	Stock Auto-Deduction	On payment, deducts ingredient stock for products in the paid order (BR-08).
4	Payment	Payment Gateway Integration	API integration submitting electronic payment requests and recording the authorization result.
5	Loyalty	Loyalty Points Accrual / Redemption	Within the Payment flow, awards loyalty points to the customer when a payment completes and applies any redeemed points.

3.1.5 Entity Relationship Diagram



Entities Description

#	Entity	Description
1	User	A system user (admin, manager, cashier, barista) with login credentials and a role.
2	Category	A grouping of products (Coffee, Tea, Pastry...).
3	Product	A menu item with name, category, price and availability.
4	Order	A customer order with status, table, customer, creator and timestamp.
5	Order Item	A line in an order: product, quantity, options, line price.
6	Payment	Payment for an order: method, amount, points redeemed, timestamp.

7	Table (TABLE_T)	A physical table with number, capacity and occupancy status.
8	Ingredient	A stock item with quantity on hand and reorder threshold.
9	Customer	A customer with contact info and loyalty points.
10	Loyalty Transaction	A record of points earned/redeemed, linked to a customer and payment.
11	Purchase Order	An order to a supplier, created by a user, with status and total.
12	Stock-In	A goods-receipt line recording received ingredient quantity against a purchase order.

3.1.6 Entity Details

1. User

#	Attribute	PK	Type	Mandatory	Description
1	UserID	x	Integer	Yes	Unique identifier.
2	Username		String	Yes	Login name; unique.
3	PasswordHash		String	Yes	Hashed password.
4	Role		String	Yes	Administrator / Manager / Cashier / Barista.
5	IsActive		Boolean	Yes	Whether the account is active.

2. Category

#	Attribute	PK	Type	Mandatory	Description
1	CategoryID	x	Integer	Yes	Unique identifier.
2	Name		String	Yes	Display name.

3. Product

#	Attribute	PK	Type	Mandatory	Description
1	ProductID	x	Integer	Yes	Unique identifier.
2	Name		String	Yes	Display name.
3	CategoryID	FK	Integer	Yes	FK to Category.
4	Price		Decimal	Yes	Unit selling price.
5	Size		String	No	Size variants (S/M/L).
6	IsAvailable		Boolean	Yes	Available for ordering.
7	Description		String	No	Optional description.

4. Order

#	Attribute	PK	Type	Mandatory	Description
1	OrderID	x	Integer	Yes	Unique identifier.
2	createBy	FK	Integer	Yes	FK to User (creator).
3	TableID	FK	Integer	No	FK to Table; null for takeaway.
4	CustomerID	FK	Integer	No	FK to Customer.
5	Status		String	Yes	Open / Paid / Cancelled.
6	CreatedAt		Datetime	Yes	Creation timestamp.

5. Order Item

#	Attribute	PK	Type	Mandatory	Description
1	OrderItemID	x	Integer	Yes	Unique identifier.
2	OrderID	FK	Integer	Yes	FK to Order.
3	ProductID	FK	Integer	Yes	FK to Product.
4	Quantity		Integer	Yes	Number of units.

5	Options		String	No	Modifiers (size, sugar level, notes).
6	LinePrice		Decimal	Yes	Line total.

6. Payment

#	Attribute	PK	Type	Mandatory	Description
1	PaymentID	x	Integer	Yes	Unique identifier.
2	OrderID	FK	Integer	Yes	FK to Order.
3	UserID	FK	Integer	Yes	FK to User (who processed it).
4	CustomerID	FK	Integer	No	FK to Customer.
5	Method		String	Yes	Cash / Card / E-Wallet.
6	Amount		Decimal	Yes	Amount paid.
7	PointsRedeemed		Integer	No	Loyalty points redeemed.
8	PaidAt		Datetime	Yes	Completion timestamp.

7. Table (TABLE_T)

#	Attribute	PK	Type	Mandatory	Description
1	TableID	x	Integer	Yes	Unique identifier.
2	Number		Integer	Yes	Table number.
3	Capacity		Integer	Yes	Number of seats.
4	OccupancyStatus		String	Yes	Free / Occupied / Reserved.
5			String	Yes	

6	Shape		String	Yes	Round/Square/
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8. Ingredient

#	Attribute	PK	Type	Mandatory	Description
1	IngredientID	x	Integer	Yes	Unique identifier.
2	Name		String	Yes	Display name.
3	QuantityOnHand		Decimal	Yes	Current stock level.
4	ReorderThreshold		Decimal	Yes	Low-stock alert level.

9. Customer

#	Attribute	PK	Type	Mandatory	Description
1	CustomerID	x	Integer	Yes	Unique identifier.
2	FullName		String	Yes	Customer name.
3	Phone		String	No	Phone number.
4	Email		String	No	Email address.
5	LoyaltyPoints		Integer	Yes	Current loyalty points balance.
6	JoinedAt		Datetime	Yes	Registration timestamp.

10. Loyalty Transaction

#	Attribute	PK	Type	Mandatory	Description
1	LoyaltyTxnID	x	Integer	Yes	Unique identifier.
2	CustomerID	FK	Integer	Yes	FK to Customer.
3	PaymentID	FK	Integer	Yes	FK to Payment.
4	Type		String	Yes	Earn / Redeem.

5	Points		Integer	Yes	Points earned or redeemed.
6	CreatedAt		Datetime	Yes	Timestamp.

11. Purchase Order

#	Attribute	PK	Type	Mandatory	Description
1	PurchaseOrderID	x	Integer	Yes	Unique identifier.
2	UserID	FK	Integer	Yes	FK to User (creator).
3	SupplierName		String	Yes	Supplier name.
4	Status		String	Yes	Order status.
5	TotalAmount		Decimal	Yes	Total cost.
6	CreatedAt		Datetime	Yes	Timestamp.

12. Stock-In

#	Attribute	PK	Type	Mandatory	Description
1	StockInID	x	Integer	Yes	Unique identifier.
2	PurchaseOrderID	FK	Integer	Yes	FK to Purchase Order.
3	IngredientID	FK	Integer	Yes	FK to Ingredient.
4	Quantity		Decimal	Yes	Quantity received.
5	UnitCost		Decimal	Yes	Cost per unit.
6	ReceivedAt		Datetime	Yes	Receipt timestamp.

3.2 Authentication

3.2.1 FUNC-01 – Login / UC01 Login

3.2.1.1 Screen Mock-up

9:41



Brew & Co.
COUNTER POS

Username

 cashier.linh

Password

Login

Trouble signing in? Ask your manager.

CSMS v2.0 · Store #HCM-01

Figure 3-1: Screen Design of Login

Table 3-1: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	User Name	Text	Yes	20	
2	Password	Text	Yes	20	Display "*" instead of clear character
3	Login	Button			Go to home page

3.2.1.2 Use Case Description

UC ID and Name:	UC01 - Login
Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Administrator, Manager, Cashier, Barista
Secondary Actors:	None
Summary:	Allows registered users to authenticate with the system using their credentials to gain access to features permitted by their assigned role.
Preconditions:	The user has a valid account registered in the system.
Postconditions:	The user is authenticated and a session is created. The system displays the home screen corresponding to the user's role.
Dependency:	None
Description of main sequence:	1. User opens the application and the system displays the login screen. 2. User enters their username and password. 3. User selects the "Login" button. 4. System validates the credentials against the database. 5. System creates a session and redirects the user to the role-based dashboard.
Description of alternative sequences:	None

Exceptions:	E1: Invalid Credentials: If the username or password is incorrect, the system displays the message "Invalid username or password" and requests re-entry. E2: Empty Fields: If a required field is left blank, the system highlights the field and prompts the user to complete it.
Nonfunctional requirements:	Security: Passwords must be transmitted and stored in encrypted form. Performance: Authentication must complete within 2 seconds.

3.2.2 FUNC-02 – Logout / UC11 Logout

3.2.2.1 Screen Mock-up

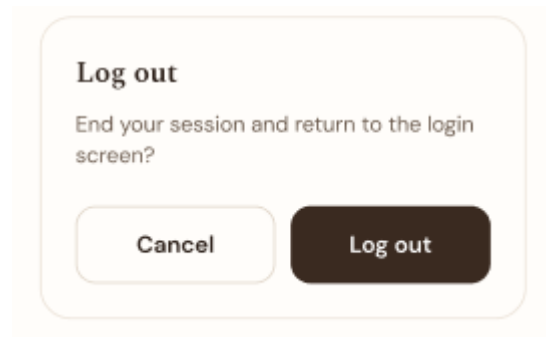


Table 3-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Delete Confirmation Message	Label	No	100	Display the message "Are you sure you want to delete this item?"
2	Cancel Delete	Button	No		Close delete confirmation
3	Delete	Button	No		Delete the selected item and close the dialog.
4	Logout Confirmation Message		No	100	Display the message "End your session and return to the login screen?"
5	Cancel Logout	Button	No		Close the logout confirmation
6	Log out	Button	No		Terminate the current session and navigate to the login screen.

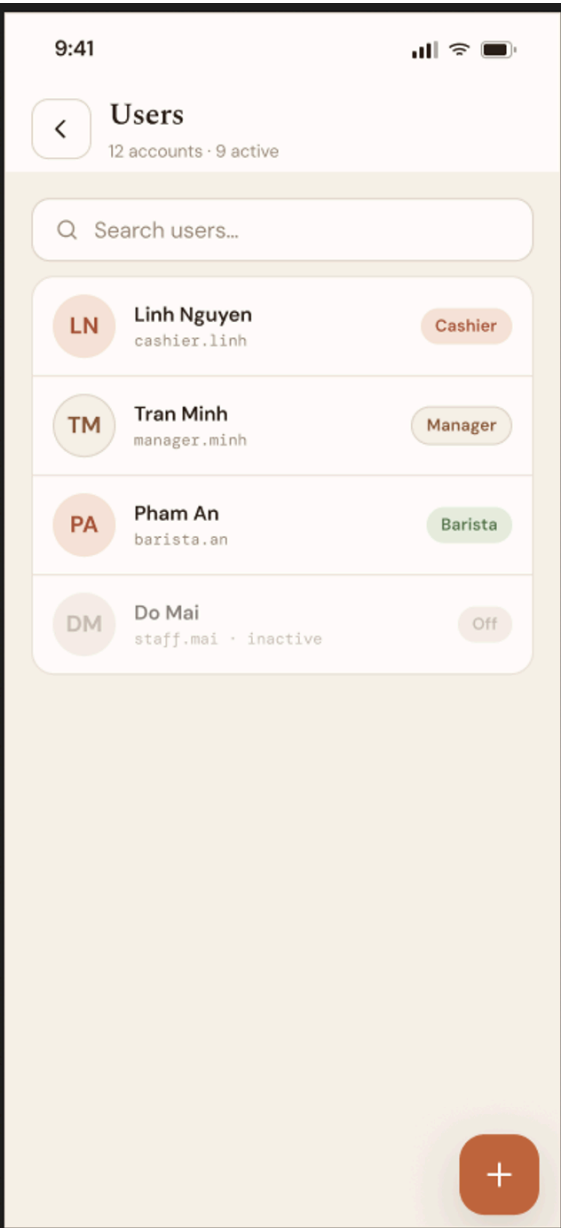
3.2.2.2 Use Case Description

UC ID and Name:	UC02 - Logout
Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Administrator, Manager, Cashier, Barista
Secondary Actors:	None
Summary:	Allows an authenticated user to safely terminate their current session and exit the system.
Preconditions:	The user is currently logged into the system.
Postconditions:	The user's session is terminated. The system returns to the login screen.
Dependency:	None
Description of main sequence:	1. User selects the "Logout" option from the menu. 2. System requests confirmation of the logout action. 3. User confirms the action. 4. System terminates the active session and clears session data. 5. System redirects the user to the login screen.
Description of alternative sequences:	A1: Cancel Logout: At step 3, if the user cancels, the system closes the confirmation dialog and keeps the session active.
Exceptions:	None
Nonfunctional requirements:	Security: All session tokens must be invalidated upon logout. Performance: Logout must complete within 2 seconds.

3.3 User Administration

3.3.1 FUNC-01/UC04: User Management

3.3.1.1 Screen Mock-Up



#	Field Name	Type	Mandatory	Max Length	Description

1	Back Button	Button	No		Navigate back to the previous dashboard screen.
2	Screen Title	Label	No	50	Display the static screen title “Users”.
3	Accounts Summary	Label	No	100	Display the short summary of total and active accounts (e.g., “12 accounts · 9 active”).
4	Search users Input	Text Input	No	100	Allow searching users by typing keywords or usernames. Placeholder: “Search users...”.
5	User List Container	Container	No		Scrollable area displaying the list card of all accounts in the system.
6	User Avatar Avatar	Image / Icon	No		Circular badge displaying the user's initials (e.g., “LN”, “TM”, “PA”, “DM”) based on their full name. Greyed out if inactive.
7	User Name Label	Label	No	50	Display the full name of the user (e.g., “Linh

					Nguyen”, “Tran Minh”, “Pham An”, “Do Mai”).
8	Username & Status Text	Label	No	100	Display the unique system handle (e.g., “cashier.linh”, “manager.minh”). Append “ · inactive” if the account is deactivated.
9	Role Badge	Label	No	20	Display a colored chip indicating the assigned system role (e.g., “Cashier”, “Manager”, “Barista”). Displays “Off” in grey if the user is inactive.
10	User Item Row	Container	No		Trigger an event when clicked, navigating the Administrator to the corresponding Edit User screen.
11	Add User Button	Button	No		Floating Action Button (FAB) positioned at the bottom right with a “+” icon to navigate to the Add User screen.

9:41

×

Add User

Full name

Linh Nguyen

Username

cashier.linh

Role

Administrator

Manager

Cashier

Barista

Account active

Can sign in to CSMS

Add User

#	Field Name	Type	Mandatory	Max Length	Description
1	Close (X)	Button	-	-	Cancels the current operation and navigates back to the User Management list screen.
2	Delete	Icon/Button	-	-	Clears all currently entered data from the form fields.
3	Full name	Text Input	Yes	50	Enters the full name of the employee being granted the account.
4	Username	Text Input	Yes	30	Enters the unique identifier used to log into the system. Displays a user icon at the start of the field.
5	Role	Radio Button Group	Yes	-	Selects a single permission level for the account
6	Account active	Toggle Switch	Yes	-	Sets the activation status of the account
7	Add User	Button	-	-	Validates data entry. Creates the account and triggers a success message upon completion.

9:41



Edit User



Full name

Linh Nguyen

Username

 cashier.linh

Role

Administrator



Manager



Cashier



Barista



Account active

Can sign in to CSMS



Save changes

#	Field Name	Type	Mandatory	Max Length	Description
1	Delete Confirmation Message	Label	No	100	Display the message “Are you sure you want to delete this item?”
2	Cancel Delete	Button	No		Close delete confirmation
3	Delete	Button	No		Delete the selected item and close the dialog.
4	Logout Confirmation Message		No	100	Display the message “End your session and return to the login screen?”
5	Cancel Logout	Button	No		Close the logout confirmation
6	Log out	Button	No		Terminate the current session and navigate to the login screen.

3.3.1.2 Use Case Description

UC ID and Name:	UC03 - Manage User
-----------------	--------------------

Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Administrator
Secondary Actors:	None
Summary:	Allows the administrator to create, view, update, and deactivate user accounts and assign roles to control system access.
Preconditions:	The administrator is successfully logged into the system.
Postconditions:	User account changes are successfully saved in the database. The system displays a confirmation message.
Dependency:	<<include>> Authorize step 1
Description of main sequence:	1. Administrator selects the "Manage User" section from the menu. 2. System retrieves and displays the list of existing user accounts. 3. Administrator selects an action (Create, Edit, or Deactivate). 4. Administrator enters or updates the account details and assigns a role. 5. Administrator submits the form. 6. System validates the data (e.g., unique username, valid role). 7. System updates the database and displays the message "User Saved Successfully".
Description of alternative sequences:	A1: Deactivate Account: At step 3, if the administrator chooses to deactivate, the system disables the account instead of deleting it and preserves historical records.
Exceptions:	E1: Duplicate Username: If the username already exists, the system highlights the field and requests a different value. E2: Invalid Data Format: If any field has an incorrect format, the system highlights it and requests a correction.
Nonfunctional requirements:	Security: Only users with the Administrator role may access this function. Performance: User records must be retrieved and displayed within 2 seconds.

3.4 Payment

3.4.1 FUNC-04-Process Payment / UC10: Process Payment

3.4.1.1 Screen Mock-up

9:41

< Payment
#ORD-1182 · T-05

Amount due
205,000đ

Payment method

☒ Cash Card E-Wallet

ML Mai Le
Balance 1,240 pts +20 pts

Redeem 500 pts
-50,000đ off total - 500 +

Subtotal	225,000đ
Discount (loyalty)	-50,000đ
Cash tendered	300,000đ
Change due	125,000đ

Confirm payment · 175,000đ

Table 3-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
---	------------	------	-----------	------------	-------------

1	Amount due	Label			Displays the total amount the customer needs to pay.
2	Payment method	Button Group	Yes		Allows selection of payment method: Cash, Card, E-Wallet.
3	Redeem pts	Numeric Input	No		Input field for loyalty points to redeem (includes - / + buttons).
4	Discount	Label			Displays the applied discount percentage and amount.
5	Cash tendered	Numeric Input	Yes		Input field for the amount of cash received (required if Cash is selected).
6	Request approval	Button			Sends an approval request to the Manager (appears if discount > 50%).
7	Confirm payment	Button			Confirms the payment and completes the order.
8	Print receipt	Button			Prints the order receipt (shown on success screen).
9	New order	Button			Navigates back to create a new order (shown on success screen).

3.4.1.2 Use Case Description

UC ID and Name:	UC17 - Process Payment
Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Cashier
Secondary Actors:	Payment Gateway
Summary:	Allows the cashier to process payment for an order using cash or an electronic method, completing the sale and generating a receipt.
Preconditions:	The cashier is logged in and an order awaiting payment exists.

Postconditions:	The payment is recorded and the order status is set to "Completed". A receipt is generated for the customer.
Dependency:	<<include>> Authorize step 1
Description of main sequence:	1. Cashier selects an order and chooses the "Process Payment" option. 2. System displays the order total and available payment methods. 3. Cashier selects a payment method (e.g., cash or card). 4. For electronic payment, the system sends the transaction request to the Payment Gateway. 5. Payment Gateway processes the transaction and returns the result. 6. System records the payment, marks the order "Completed", and generates a receipt.
Description of alternative sequences:	A1: Cash Payment: At step 3, if cash is selected, the system calculates the change and skips the Payment Gateway steps.
Exceptions:	E1: Payment Declined: If the Payment Gateway rejects the transaction, the system notifies the cashier and requests an alternative method. E2: Gateway Unavailable: If the Payment Gateway cannot be reached, the system displays an error and allows retry or cash payment.
Nonfunctional requirements:	Security: Payment transactions must be transmitted securely and no card data is stored locally. Performance: Payment confirmation must complete within 5 seconds.

3.5 Order Management

3.5.1 FUNC-05-Create Order/ UC6: Create Order

3.5.1.1 Screen Mock-up



Table 3-3: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Back	Button			Returns to the previous screen without saving the order.
2	Cancel	Button			Discards the current new order and exits the screen.
3	Table Selector	Button	No		Opens table selection to assign a table and number of seats (e.g., T-05 · 4 seats).
4	Add Customer	Button	No		Attaches an existing customer to the order for loyalty tracking.
5	Search Menu	Text Input	No	50	Searches the menu for products by name.
6	Category Tabs	Tab / Button	No		Filters products by category (Coffee, Tea, Pastry, Cold).
7	Product Card	Container	No		Displays a product with name, price and size options (e.g.,

					S/M/L); tapping adds it to the order.
8	Order Summary	Label			Shows the running item count and total amount of the order.
9	Review Order	Button			Proceeds to the order review and confirmation step.

3.5.1.2 Use Case Description

UC ID and Name:	UC14 - Create Order
Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Cashier
Secondary Actors:	None
Summary:	Allows the cashier to create a new customer order by selecting products, quantities, and an optional table, initiating the sales process.
Preconditions:	The cashier is successfully logged into the system and products are available.
Postconditions:	A new order is created and saved in the database with status "Pending". Ingredient inventory is deducted according to the ordered products. The order appears in the order queue.
Dependency:	<<include>> Authorize step 1, <<include>> Process payment step 7
Description of main sequence:	1. Cashier selects the "Create Order" section from the menu. 2. System displays the product menu and an empty order. 3. Cashier selects products and specifies quantities. 4. Cashier optionally assigns a table and a customer to the order. 5. Cashier submits the order. 6. System validates the order (e.g., at least one item, sufficient ingredient stock). 7. System saves the order, deducts ingredient inventory, and forwards the order to payment.

Description of alternative sequences:	A1: Add Customer: At step 4, if the customer is not registered, the cashier may create a new customer record.
Exceptions:	E1: Empty Order: If no products are selected, the system prevents submission and prompts the cashier to add items. E2: Insufficient Stock: If an ingredient is out of stock, the system alerts the cashier and prevents the order until resolved.
Nonfunctional requirements:	Security: Only authorized roles may create orders. Performance: The order must be saved within 2 seconds.

3.5.2 *FUNC-05*-Order Management/ *UC7*: Update Order

3.5.2.1 Screen Mock-up

9:41

Coffee

Tea

C

Cappuccino

45,000đ

Size

S · 40k

M · 45k

L · 52k

Sugar level

0%

30%

50%

100%

Notes optional

e.g. extra hot, oat milk...

Quantity

—

2

+

Add 2 · 90,000đ

Table 3-4: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Product Name	Label			Displays the selected product name (e.g., Cappuccino).
2	Product Price	Label			Displays the unit price of the product (e.g., 45,000đ).
3	Size	Radio Button Group	Yes		Selects one size option, each with its own price (e.g., S·40k, M·45k, L·52k).
4	Sugar Level	Radio Button Group	Yes		Selects the sugar level (0%, 30%, 50%, 100%).
5	Notes	Text Input	No	200	Optional free-text note for the item (e.g., extra hot, oat milk).
6	Quantity	Numeric Input	Yes		Sets the item quantity using the – / + stepper (minimum 1).
7	Add to Order	Button			Adds the item with the selected options to the order; the button shows the quantity

					and line total (e.g., Add 2 · 90,000đ).
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3.5.2.2 Use Case Description

UC ID and Name:	UC15 - Update Order
Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Cashier
Secondary Actors:	None
Summary:	Allows the cashier to modify an existing pending order by adding, removing, or changing items before payment is completed.
Preconditions:	The cashier is logged in and a pending order exists.
Postconditions:	The updated order is saved in the database. Ingredient inventory is adjusted to reflect the changes. The system displays a confirmation message.
Dependency:	<<include>> Authorize step 1
Description of main sequence:	1. Cashier selects an existing order from the order queue. 2. System retrieves and displays the order details. 3. Cashier selects the "Edit Order" option. 4. Cashier adds, removes, or changes the quantity of items. 5. Cashier submits the changes. 6. System validates the updated order (e.g., sufficient ingredient stock). 7. System updates the order, adjusts inventory, and displays the message "Order Updated Successfully".
Description of alternative sequences:	None
Exceptions:	E1: Order Already Paid: If the order has already been paid, the system prevents editing and notifies the cashier. E2: Insufficient Stock: If an added item exceeds available stock, the system alerts the cashier.
Nonfunctional requirements:	Security: Only authorized roles may update orders. Performance: Order updates must be confirmed within 2 seconds.

3.5.3 FUNC-05-Order Management/ UC8: Cancel Order

3.5.3.1 Screen Mock-up

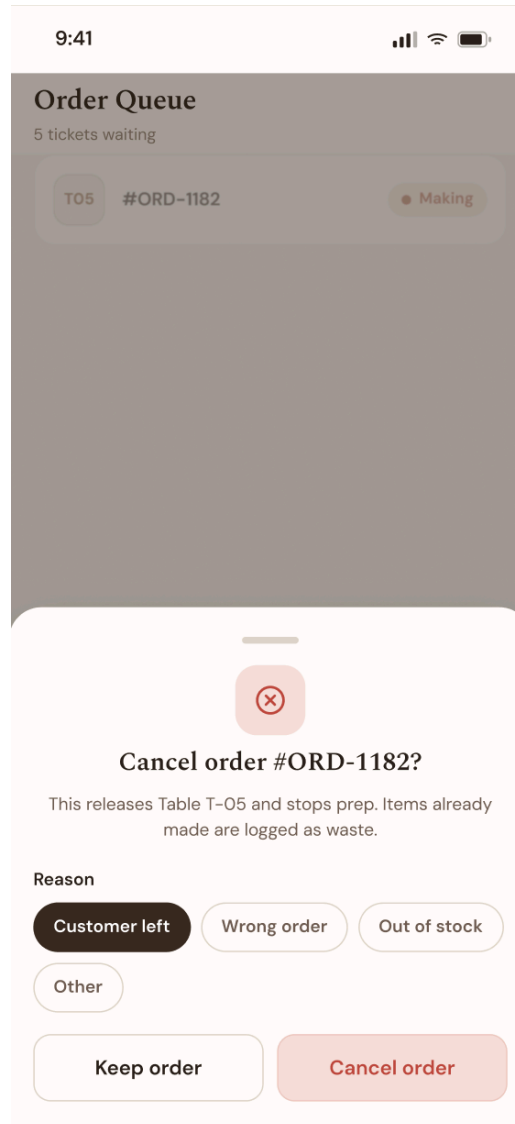


Table 3-5: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Close (X)	Button			Dismisses the dialog without cancelling the order.
2	Confirmation Message	Label			Displays the cancel

					confirmation text including the order ID and impact (e.g., releases the table and logs prepared items as waste).
3	Reason	Radio Button Group	Yes		Selects the cancellation reason (Customer left, Wrong order, Out of stock, Other).
4	Keep Order	Button			Closes the dialog and keeps the order unchanged.
5	Cancel Order	Button			Confirms cancellation and sets the order status to Cancelled.

3.5.3.2 Use Case Description

UC ID and Name:	UC16 - Cancel Order
Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Cashier
Secondary Actors:	None
Summary:	Allows the cashier to cancel an existing pending order, releasing any assigned table and restoring deducted ingredient inventory.
Preconditions:	The cashier is logged in and a pending order exists.
Postconditions:	The order status is set to "Cancelled" in the database. Deducted ingredient inventory is restored and any assigned table is released. The system displays a confirmation message.

Dependency:	<<include>> Authorize step 1
Description of main sequence:	1. Cashier selects an existing order from the order queue. 2. System retrieves and displays the order details. 3. Cashier selects the "Cancel Order" option. 4. System requests confirmation and, optionally, a cancellation reason. 5. Cashier confirms the cancellation. 6. System sets the order to "Cancelled", restores inventory, releases the table, and displays the message "Order Cancelled Successfully".
Description of alternative sequences:	A1: Cancel Aborted: At step 5, if the cashier declines, the system closes the dialog and keeps the order unchanged.
Exceptions:	E1: Order Already Paid: If the order has already been paid, the system prevents cancellation and notifies the cashier.
Nonfunctional requirements:	Security: Only authorized roles may cancel orders. Performance: Cancellation must be processed within 2 seconds.

3.5.3 FUNC-05-Order Management/ UC9: Assign Table

3.5.3.1 Screen Mock-up

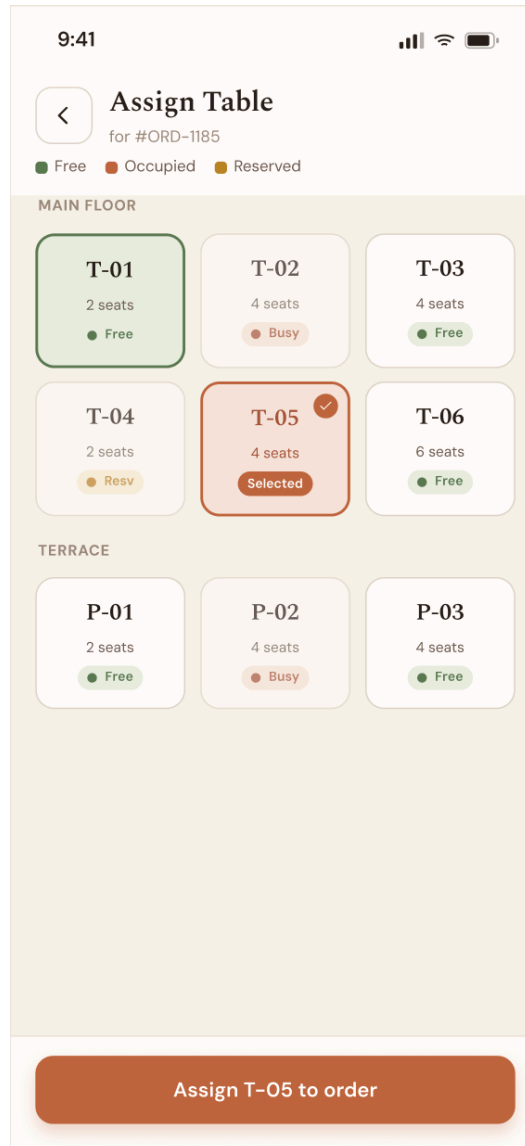


Table 3-6: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Back	Button			Returns to the previous screen without assigning a table.

2	Screen Title	Label			Displays the screen title and the order ID being assigned (e.g., for #ORD-1185).
3	Status Legend	Label			Shows the colour legend for table status (Free, Occupied, Reserved).
4	Floor Section	Label			Groups tables by area (Main Floor, Terrace).
5	Table Card	Container	No		Displays a table with its code, seat count and status; a free table can be tapped to select it (e.g., T-05).
6	Assign Button	Button			Links the selected table to the order and sets the table status to Occupied (e.g., Assign T-05 to order).

3.6 Product Management

3.6.1 Func 09- Manage Product UC13

3.6.1.1 Screen Mockup

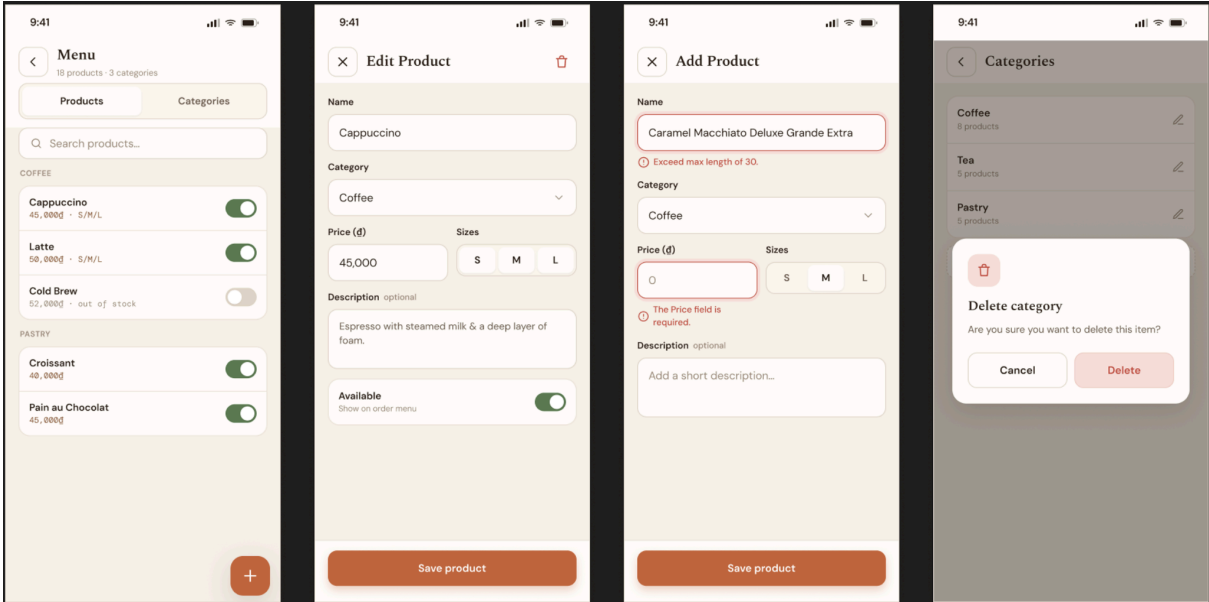


Table 3-3: Screen Definition — View Product

#	Field Name	Type	Mandatory	Max Length	Description
1	Back Button	Button	No		Navigate back to the previous dashboard screen.
2	Screen Title	Label	No	50	Display the static screen title “Menu”.
3	Menu Summary	Label	No	100	Display the short summary of items (e.g., “18 products · 3 categories”).

4	Products Tab	Tab / Button	No		Switch the list view to display all products (Active state).
5	Categories Tab	Tab / Button	No		Switch the list view to display product categories.
6	Search products Input	Text Input	No	100	Allow searching products by typing keywords.
7	Category Header	Label	No	50	Display the category name grouping the products (e.g., “COFFEE”, “PASTRY”).
8	Product Item Row	Container	No		Display product details including Name, Price, and available Sizes (or status like “out of stock”). Clicking the row triggers the Edit view.
9	Product Status Toggle	Toggle Switch	Yes		Quick enable or disable the product availability. Green indicates active status.

10	Add Product Button	Button	No		Floating Action Button (FAB) with a “+” icon to navigate to the Add Product screen.
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Table 3-4: Screen Definition — Edit Product

#	Field Name	Type	Mandatory	Max Length	Description
1	Close (X)	Button	No		Cancel the current operation and navigate back to the Menu Management list screen.
2	Screen Title	Label	No	50	Display the static screen title “Edit Product”.
3	Delete Icon	Button	No		Trigger the delete confirmation popup (MSG06) for this specific product.
4	Name Label	Label	No	30	Display the text label “Name”.
5	Name Input	Text Input	Yes	30	Enter or edit the product name. Default text: “Cappuccino”.

6	Category Label	Label	No	50	Display the text label “Category”.
7	Category Dropdown	Dropdown	Yes		Select a product category from the list. Current value: “Coffee”.
8	Price (₫) Label	Label	No	20	Display the text label “Price (₫)”.
9	Price Input	Text Input	Yes	15	Enter the price of the product. Displays value with comma separating thousands (e.g., “45,000”).
10	Sizes Label	Label	No	20	Display the text label “Sizes”.
11	Size Option S	Button / Checkbox	Yes		Toggle selection for size “S”.
12	Size Option M	Button / Checkbox	Yes		Toggle selection for size “M”.
13	Size Option L	Button / Checkbox	Yes		Toggle selection for size “L”.

14	Description Label	Label	No	30	Display the text label “Description” with optional hint.
15	Description Input	Text Input	No	255	Enter detailed product notes. Default: “Espresso with steamed milk & a deep layer of foam.”.
16	Available Setting Row	Label	No	50	Display setting row labels “Available” and description “Show on order menu”.
17	Available Toggle	Toggle Switch	Yes		Enable or disable the product availability. Default: Active (Green).
18	Save product	Button	No		Validate all input fields, save the changes, and show success message MSG05.

Table 3-5: Screen Definition — Add Product Validation (Màn hình 3)

#	Field Name	Type	Mandatory	Max Length	Description

1	Close (X)	Button	No		Cancel the product creation process and navigate back to the list screen.
2	Screen Title	Label	No	50	Display the static screen title “Add Product”.
3	Name Label	Label	No	30	Display the text label “Name”.
4	Name Input	Text Input	Yes	30	Input a new product name. Border turns red on error. Current input: “Caramel Macchiato Deluxe Grande Extra”.
5	Name Error Message	Label	No	50	Inline validation error text: “Exceed max length of 30.”.
6	Category Label	Label	No	50	Display the text label “Category”.
7	Category Dropdown	Dropdown	Yes		Select a product category from the list. Default: “Coffee”.

8	Price (₫) Label	Label	No	20	Display the text label “Price (₫)”.
9	Price Input	Text Input	Yes	15	Input the pricing value. Border turns red on empty field error. Current text: “0”.
10	Price Error Message	Label	No	50	Inline validation error text: “The Price field is required.”.
11	Sizes Label	Label	No	20	Display the text label “Sizes”.
12	Size S / M / L	Button Group	Yes		Select the supported sizes for the new product. Size “M” is active in the image.
13	Description Label	Label	No	30	Display the text label “Description optional”.
14	Description Placeholder	Text Input	No	255	Display helper placeholder text: “Add a short description...”.

15	Save product	Button	No		Execute field validations. Block operation until validation errors are resolved.
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Table 3-6: Screen Definition — Delete Product

#	Field Name	Type	Mandatory	Max Length	Description
1	Back Button	Button	No		Navigate back to the previous screen.
2	Screen Title	Label	No	50	Display the static screen title “Categories”.
3	Category List Items	Label	No	50	Display list elements containing category name and product amount count (e.g., “Coffee, 8 products”).
4	Edit Category Icon	Button	No		Trigger editing the selected category item line.
5	Dialog Overlay Background	Container	No		Darkened background blocking interactions with the background categories screen.

6	Dialog Alert Trash Icon	Icon	No		Red alert decorative trash icon at the top left of the dialog form.
7	Dialog Box Title	Label	No	50	Display confirmation card header text “Delete category”.
8	Delete Confirmation Message	Label	No	100	Display the message “Are you sure you want to delete this item?”.
9	Cancel	Button	No		Close delete confirmation dialog without modifications.
10	Delete	Button	No		Confirm deactivating/deleting the category, trigger success workflow, and close the dialog

3.6.1.2 Use case description

UC ID and Name:	UC04 - Manage Product
Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Manager
Secondary Actors:	None
Summary:	Allows the manager to add, edit, and remove products in the menu, including price, description, and category assignment.
Preconditions:	The manager is successfully logged into the system.

Postconditions:	Product information is successfully saved in the database. The system displays a confirmation message.
Dependency:	<<include>> Authorize step 1
Description of main sequence:	1. Manager selects the "Manage Product" section from the menu. 2. System retrieves and displays the current list of products. 3. Manager selects an action (Add, Edit, or Delete). 4. Manager enters or updates the product details (name, price, category, image). 5. Manager submits the form. 6. System validates the data (e.g., price is a positive number, required fields present). 7. System updates the database and displays the message "Product Saved Successfully".
Description of alternative sequences:	A1: Delete Product: At step 3, if the manager chooses to delete, the system requests confirmation before removing the product.
Exceptions:	E1: Invalid Data Format: If any field has an incorrect format (e.g., a non-numeric price), the system highlights it and requests a correction.
Nonfunctional requirements:	Security: Only authorized roles may modify product data. Performance: The product list must load within 2 seconds.

3.7 Inventory Management

3.7.1 UC17 View Inventory Report

3.7.1.1 Screen Mockup

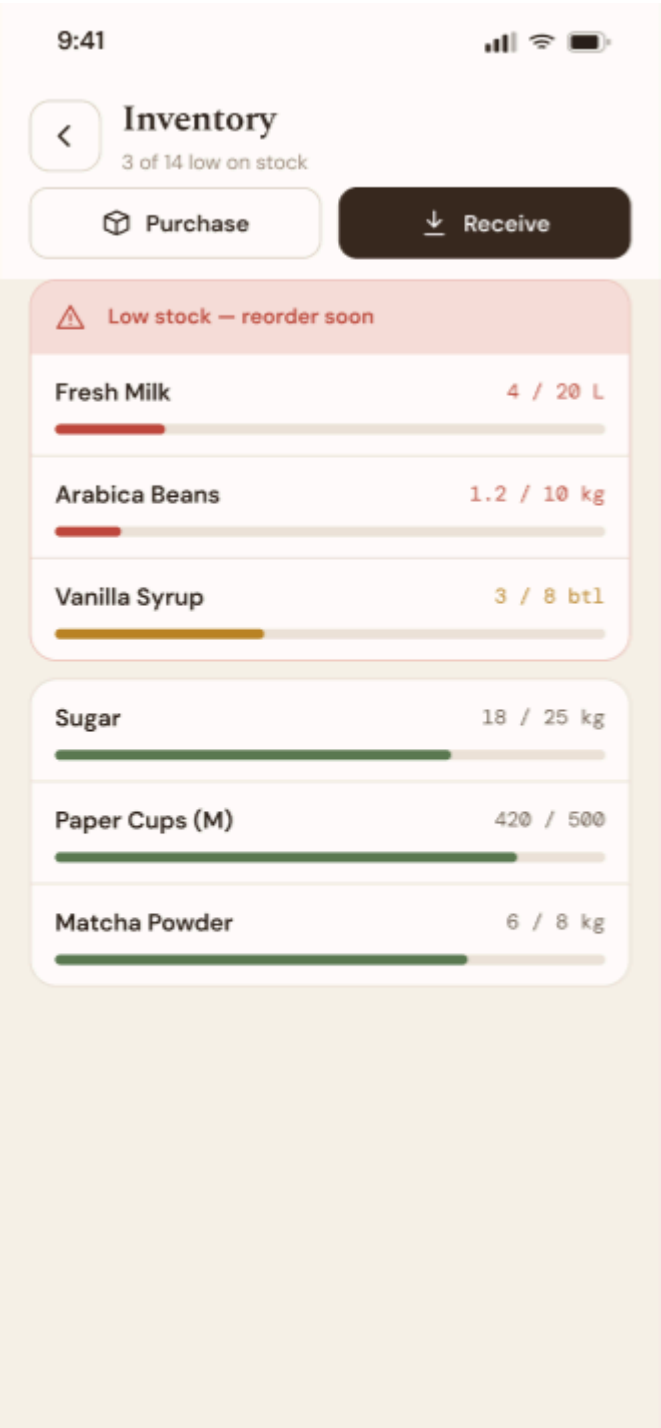


Table 3-7: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Purchase	Button	No		Opens the screen to create a new Purchase Order.
2	Receive	Button	No	50	Opens the Stock-In screen for receiving goods.
3	Low stock alert	Label	No	100	Display the short summary of items (e.g., "18 products · 3 categories").
4	Ingredient list	List	No		Displays a warning (in red) for ingredients falling below the reorder threshold.

3.7.1.2 Use case description

UC ID and Name:	UC08 - View Inventory Report
Created By:	HaiTV Date Created: 14/01/2026
Primary Actor:	Manager
Secondary Actors:	None
Summary:	Allows the manager to view current inventory levels of ingredients, including low-stock alerts, to support restocking decisions.
Preconditions:	The manager is successfully logged into the system.
Postconditions:	The current inventory report is displayed to the manager.
Dependency:	<<include>> Authorize step 1

Description of main sequence:	1. Manager selects the "View Inventory Report" section from the menu. 2. System retrieves current stock levels for all ingredients from the database. 3. System displays the inventory report, highlighting items below their stock threshold. 4. Manager may filter or sort the report as needed.
Description of alternative sequences:	A1: No Low Stock: If no items are below threshold, the system displays the report without any alerts.
Exceptions:	E1: No Data Available: If no inventory data exists, the system displays the message "No inventory records found".
Nonfunctional requirements:	Security: Only authorized roles may view inventory reports. Performance: The report must be generated within 3 seconds.

3.7.2 Func 08- Purchase Order

3.7.2.1 Screen Mock-up

9:41

<

Purchase Order

● Draft

#PO-0461 - Draft

Supplier

Saigon Coffee Supply Co. ▾

ITEMS

Arabica Beans 10 kg × 220,000đ	2,200,000đ
Fresh Milk 40 L × 28,000đ	1,120,000đ
Vanilla Syrup 12 btl × 95,000đ	1,140,000đ

+ Add line item

Total4,460,000đ

Save

Submit order

#	Field Name	Type	Mandatory	Max Length	Description
1	Supplier	Dropdown	Yes	-	Selects the ingredient supplier.
2	Items list	List	-	-	Displays the list of ordered ingredients, quantity, unit price, and line total.
3	Add line item	Button	-	-	Adds a new ingredient to the purchase order.
4	Total	Label	-	-	Displays the total cost of the purchase order.
5	Save	Button	-	-	Saves the purchase order as a Draft.
6	Submit order	Button	-	-	Confirms and submits the purchase order to the supplier.

3.7.3 UC16 Create Stock-In

3.7.3.1 Screen Mockup

9:41

<

Stock-In

Receive against #PO-0461

Saigon Coffee Supply Co.

PO-0461 · ordered Jun 4

RECEIVED QUANTITIES

Arabica Beans

Ordered 10 kg

—

10

+

Fresh Milk

Ordered 40 L

—

38

+

●

2 short of order

Vanilla Syrup

Ordered 12 btl

—

12

+

Confirm goods receipt

#	Field Name	Type	Mandatory	Max Length	Description
1	PO Info	Label			Displays the supplier's name and the referenced Purchase Order ID.
2	Received Quantities	List			Lists the ordered ingredients and the expected quantities.
3	Quantity Input	Numeric Input	Yes		Allows adjusting the actual received quantity using (- / +) buttons.
4	Shortage alert	Label			Displays an alert if the actual received quantity is less than ordered (e.g., "2 short of order"). .
5	Confirm goods receipt	Button			Confirms the receipt of goods and updates the stock on hand.

3.7.3.2 Use case description

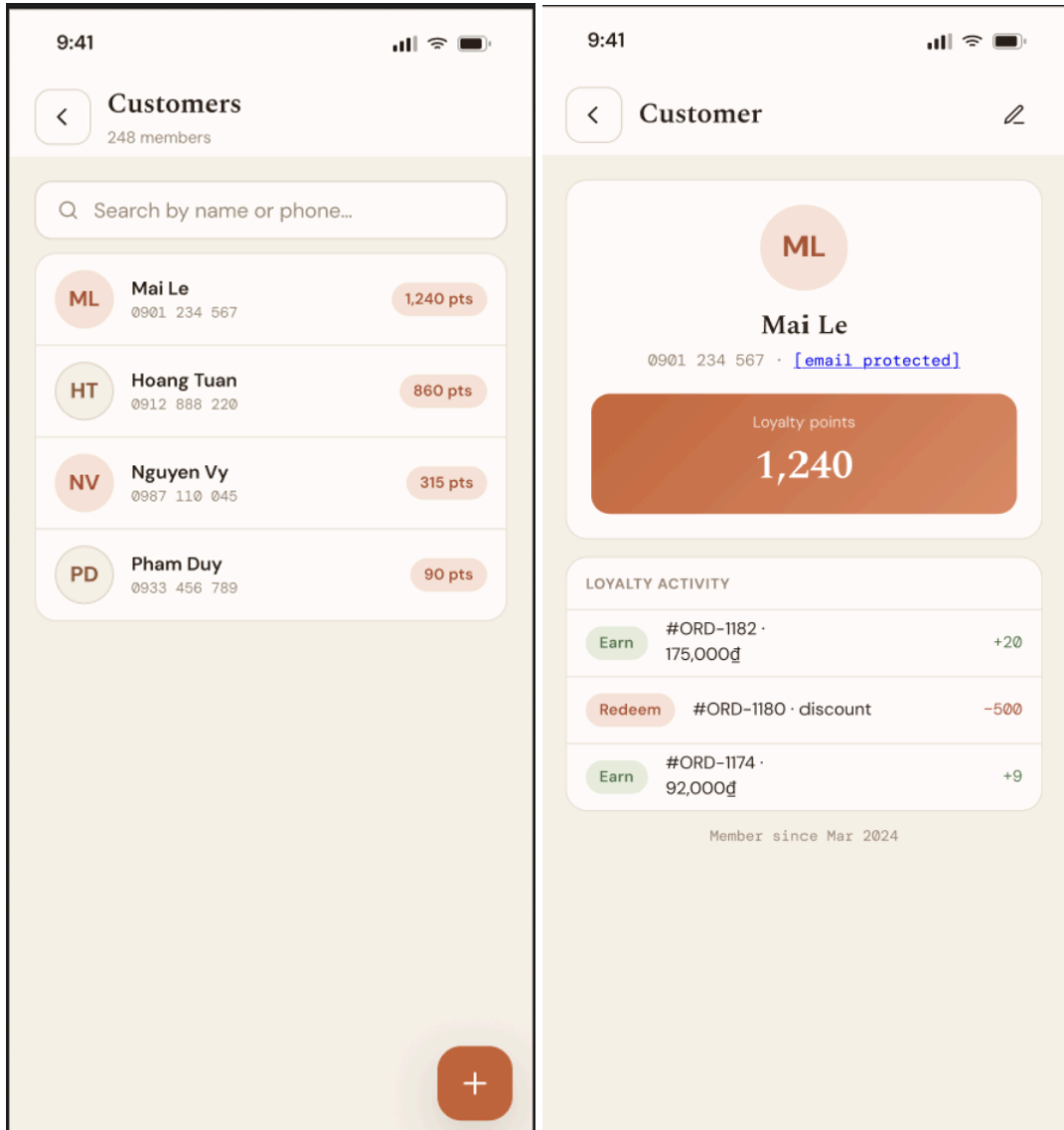
UC ID and Name:	UC08 - View Inventory Report	
Created By:	HaiTV	Date Created: 14/01/2026
Primary Actor:	Manager	

Secondary Actors:	None
Summary:	Allows the manager to view current inventory levels of ingredients, including low-stock alerts, to support restocking decisions.
Preconditions:	The manager is successfully logged into the system.
Postconditions:	The current inventory report is displayed to the manager.
Dependency:	<<include>> Authorize step 1
Description of main sequence:	1. Manager selects the "View Inventory Report" section from the menu. 2. System retrieves current stock levels for all ingredients from the database. 3. System displays the inventory report, highlighting items below their stock threshold. 4. Manager may filter or sort the report as needed.
Description of alternative sequences:	A1: No Low Stock: If no items are below threshold, the system displays the report without any alerts.
Exceptions:	E1: No Data Available: If no inventory data exists, the system displays the message "No inventory records found".
Nonfunctional requirements:	Security: Only authorized roles may view inventory reports. Performance: The report must be generated within 3 seconds.

3.8 Customer Management

3.8.1 UC19: Manage Customer

3.8.1.1 Screen Mock-up



#	Field Name	Type	Mandatory	Max Length	Description

1	Back Button	Button	No		Navigate back to the previous dashboard screen.
2	Screen Title	Label	No	50	Display the static screen title “Customers”.
3	Members Summary	Label	No	100	Display the total count of registered members (e.g., “248 members”).
4	Search Input	Text Input	No	100	Allow searching customers by entering name or phone number. Placeholder: “Search by name or phone...”.
5	Customer List Container	Container	No		Scrollable panel containing the row elements for all registered customers.
6	Customer Avatar	Image / Icon	No		Circular badge displaying the customer's initials (e.g., “ML”, “HT”, “NV”, “PD”).
7	Customer Name	Label	No	50	Display the full name of the customer (e.g., “Mai Le”, “Hoang Tuan”).
8	Phone Number	Label	No	20	Display the registered mobile number format (e.g., “0901 234 567”).

9	Loyalty Points Badge	Label	No	20	Display the total accumulated loyalty points balance (e.g., “1,240 pts”).
10	Customer Item Row	Container	No		Trigger navigation to the Customer Detail view when clicked.
11	Add Customer Button	Button	No		Floating Action Button (FAB) with a “+” icon to navigate to the Add Customer screen.

#	Field Name	Type	Mandatory	Max Length	Description
1	Back Button	Button	No		Navigate back to the Customer Management list screen.
2	Screen Title	Label	No	50	Display the static screen title “Customer”.
3	Edit Icon	Button	No		Navigate to the Edit Customer Form screen.
4	Customer Profile Badge	Image / Icon	No		Large circular badge displaying the customer's initials (“ML”).

5	Customer Name	Label	No	50	Display the detailed profile full name (“Mai Le”).
6	Contact Information	Label	No	150	Display the phone number and protected email address string.
7	Loyalty Points Card	Container	No		Highlight card section component with background fill for loyalty points display.
8	Loyalty Points Value	Label	No	20	Display the aggregate points sum value text (“1,240”).
9	Loyalty Activity Header	Label	No	50	Display section grouping header title “LOYALTY ACTIVITY”.
10	Activity History List	Container	No		List container displaying current point transactional operations.
11	Activity Type Tag	Label	No	20	Display operation type badge flag (e.g., “Earn” in green or “Redeem” in orange).

12	Activity Detail	Label	No	100	Display reference order ID and metrics (e.g., “#ORD-1182 · 175,000đ” or “#ORD-1180 · discount”).
13	Activity Points Delta	Label	No	20	Display the numeric change value with math operators (e.g., “+20” or “-500”).
14	Membership Timestamp	Label	No	50	Display chronological registration summary text

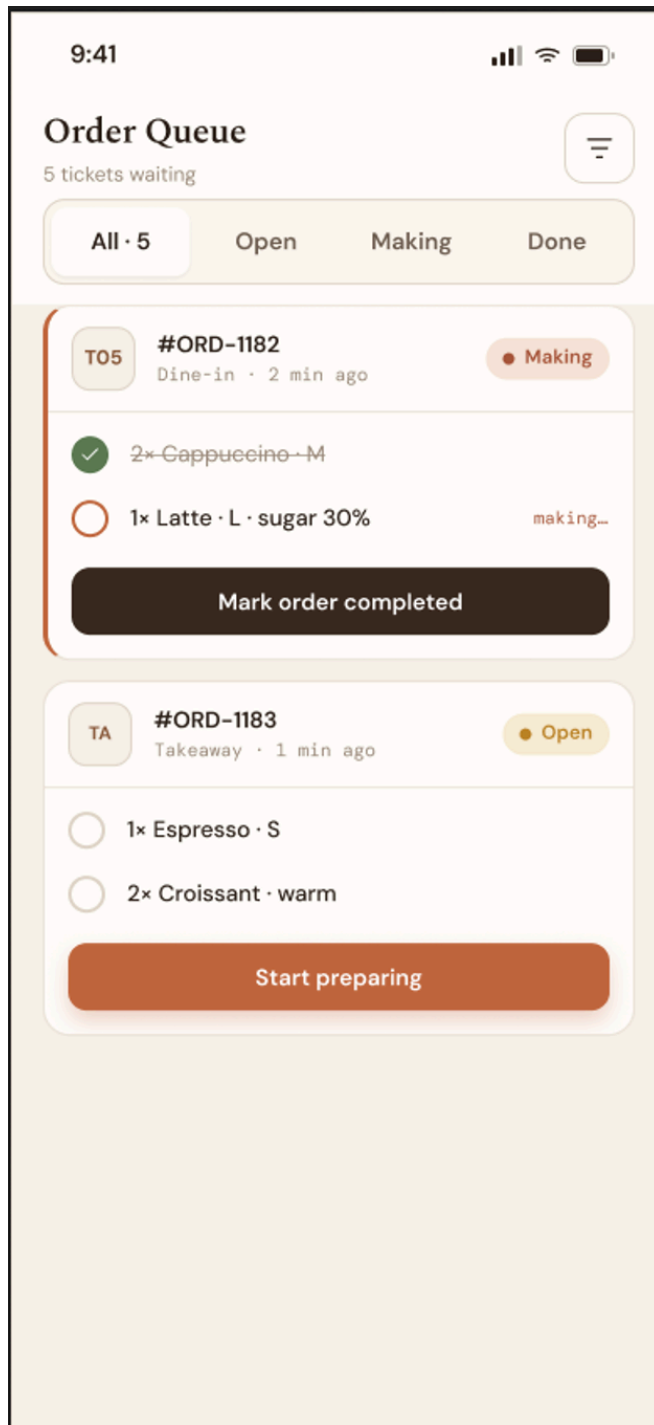
3.8.1.2 Use case description

UC ID and Name:	UC10 - Manage Customer
Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Manager
Secondary Actors:	None
Summary:	Allows the manager to add, edit, and remove customer records, including contact details and loyalty information.
Preconditions:	The manager is successfully logged into the system.
Postconditions:	Customer information is successfully saved in the database. The system displays a confirmation message.
Dependency:	<<include>> Authorize step 1
Description of main sequence:	1. Manager selects the "Manage Customer" section from the menu. 2. System retrieves and displays the list of existing customers. 3. Manager selects an action (Add, Edit, or Delete). 4. Manager enters or updates the customer details (name, phone, email). 5. Manager submits the form. 6. System validates the data (e.g., valid phone number and email format). 7. System updates the database and displays the message "Customer Saved Successfully".
Description of alternative sequences:	A1: Delete Customer: At step 3, if the manager chooses to delete, the system requests confirmation before removing the record.
Exceptions:	E1: Invalid Data Format: If the phone number or email has an incorrect format, the system highlights the field and requests a correction.
Nonfunctional requirements:	Security: Customer personal data must be protected and accessible only to authorized roles. Performance: Customer operations must be confirmed within 2 seconds.

3.9 Order Queue

3.9.1 UC11: View Order Queue

3.9.1.1 Screen Mock-up



#	Field Name	Type	Mandatory	Max Length	Description
1	Screen Title	Label	No	50	Display the static screen title “Order Queue”.
2	Queue Summary	Label	No	100	Display the short summary of waiting tickets (e.g., “5 tickets waiting”).
3	Filter Settings Button	Button	No		Icon button to open advanced filtering configurations for the order list.
4	All Tab	Tab / Button	No		Filter and display all tickets in the queue (Active state, showing “All · 5”).
5	Open Tab	Tab / Button	No		Filter and display only tickets with an Open status.
6	Making Tab	Tab / Button	No		Filter and display only tickets currently in preparation.
7	Done Tab	Tab / Button	No		Filter and display only completed tickets.

8	Ticket Card Container	Container	No		Card module displaying comprehensive details for a single order ticket.
9	Table / Service Badge	Label	No	20	Display the location identification (e.g., “T05” for Table 5 or “TA” for Takeaway).
10	Order ID Header	Label	No	30	Display the unique system reference code (e.g., “#ORD-1182”, “#ORD-1183”).
11	Service Type & Time	Label	No	100	Display fulfillment class and creation duration metadata (e.g., “Dine-in · 2 min ago” or “Takeaway · 1 min ago”).
12	Ticket Status Badge	Label	No	20	Colored indicator badge reflecting the processing phase (e.g., “Making” in orange or “Open” in yellow).

13	Item Checkbox Indicator	Checkbox / Button	No		Interactive circle indicator representing item state. Changes to a green checkmark when an individual product item is completed.
14	Order Item Line	Label	No	150	Display the requested product quantity, name, and options (e.g., “2× Cappuccino · M” with strikethrough text if done, or “1× Latte · L · sugar 30%”).
15	Individual Item Status	Label	No	20	Minor status alert text aligned to the right of an item (e.g., “making...” in orange).
16	Mark order completed	Button	No		Action button available on tickets in preparation to change the entire order status to Completed.

17	Start preparing	Button	No		Action button available on unfulfilled tickets to change the entire order status to In-progress.
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3.9.1.2 Use case description

UC ID and Name:	UC13 - View Order Queue
Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Barista, Cashier
Secondary Actors:	None
Summary:	Allows baristas and cashiers to view the queue of active orders and their preparation status in real time.
Preconditions:	The user is successfully logged into the system.
Postconditions:	The current order queue is displayed to the user.
Dependency:	<<include>> Authorize step 1
Description of main sequence:	1. User selects the "View Order Queue" section from the menu. 2. System retrieves all active orders from the database. 3. System displays the order queue ordered by time, showing each order's items and status. 4. System updates the queue automatically as order statuses change.
Description of alternative sequences:	A1: Update Status: The barista may update an order's preparation status (e.g., to "Preparing" or "Ready").
Exceptions:	E1: Empty Queue: If there are no active orders, the system displays the message "No active orders".
Nonfunctional requirements:	Security: Only authorized roles may view the order queue. Performance: The queue must refresh within 2 seconds of a status change.

3.10 Report

3.10.1 UC20: View Sales Report

3.10.1.1 Screen Mock-up

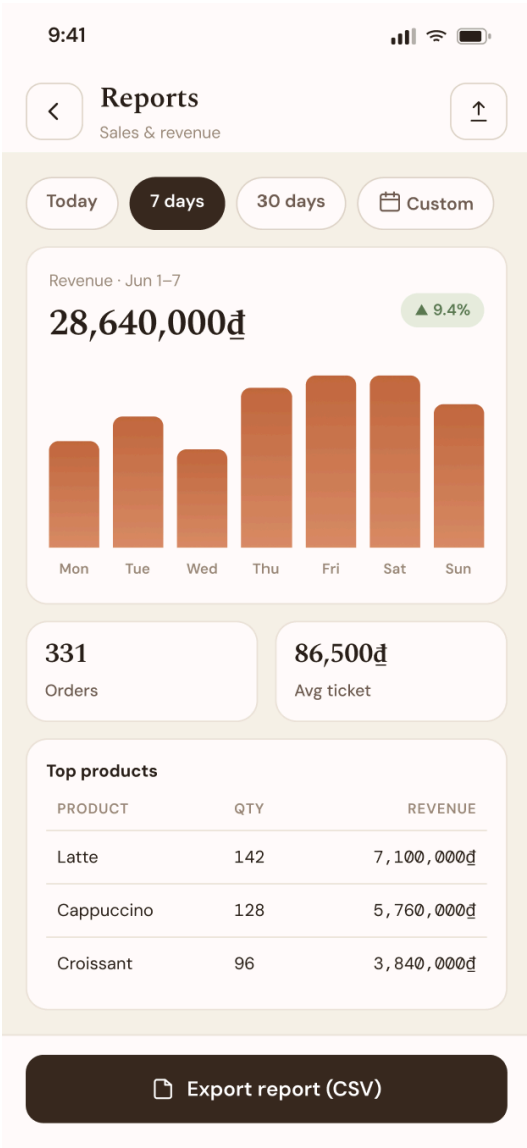


Table 3-11: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Back	Button			Returns to the previous screen.
2	Export Icon	Button			Opens export options for the current report (shortcut in the header).

3	Date Range Tabs	Tab / Button	No		Selects the reporting period (Today, 7 days, 30 days, Custom).
4	Revenue Summary	Label			Displays total revenue for the selected period with a comparison indicator (e.g., 28,640,000đ, ▲9.4%).
5	Revenue Chart	Container			Bar chart of daily revenue across the selected period (Mon–Sun).
6	Stat Cards	Label			Shows summary metrics for the period, such as number of orders and average ticket.
7	Top Products	Container			Lists best-selling products with quantity (QTY) and revenue columns.
8	Export Report (CSV)	Button			Exports the current report as a CSV file.

3.10.1.2 Use case description

UC ID and Name:	UC11 - View Sales Report
Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Manager
Secondary Actors:	None
Summary:	Allows the manager to view sales performance over a selected period, including revenue, order counts, and best-selling products.
Preconditions:	The manager is successfully logged into the system.
Postconditions:	The sales report for the selected period is displayed to the manager.
Dependency:	<<include>> Authorize step 1
Description of main sequence:	1. Manager selects the "View Sales Report" section from the menu. 2. Manager selects the reporting period (e.g., day, week, month). 3. System retrieves the sales data for the selected period from the database. 4. System calculates and displays the sales report (total revenue, number of orders, top products).
Description of alternative sequences:	A1: Custom Range: At step 2, the manager may specify a custom date range.
Exceptions:	E1: No Data Available: If no sales exist for the selected period, the system displays the message "No sales records found".
Nonfunctional requirements:	Security: Only authorized roles may view sales reports. Performance: The report must be generated within 3 seconds.

UC ID and Name:	UC12 - Export Report
Created By:	HaiVN Date Created: 14/01/2026
Primary Actor:	Manager
Secondary Actors:	None
Summary:	Allows the manager to export inventory or sales reports to an external file format (e.g., PDF or Excel) for offline use or record keeping.
Preconditions:	The manager is successfully logged into the system and a report is available.
Postconditions:	The report is exported and downloaded as a file. The system displays a confirmation message.
Dependency:	<<include>> Authorize step 1
Description of main sequence:	1. Manager selects the "Export Report" option from a displayed report. 2. System prompts the manager to choose the export format. 3. Manager selects the desired format (e.g., PDF or Excel). 4. System generates the file from the report data. 5. System provides the file for download and displays the message "Report Exported Successfully".
Description of alternative sequences:	None
Exceptions:	E1: Export Failure: If the file cannot be generated, the system displays an error message and requests the manager to retry.
Nonfunctional requirements:	Security: Only authorized roles may export reports. Performance: File generation must complete within 5 seconds.

4. Non-Functional Requirements

4.1 External Interfaces

User Interfaces: a Flutter-based mobile application on Android and iOS tablets/phones, optimized for fast counter order entry.

Hardware Interfaces: a receipt/thermal printer for receipts and kitchen tickets; an optional barcode scanner for product lookup.

Software Interfaces: an external payment gateway (REST API over HTTPS) for card and e-wallet payments.

Communication Interfaces: the app communicates with the backend over HTTPS via the cafe's Wi-Fi/internet; gateway calls also use HTTPS.

4.2 Quality Attributes

4.2.1 Usability

A new cashier should learn to create and complete an order within 30 minutes of training. A typical order of up to 5 items should be enterable in under 60 seconds. Clear labels, large touch targets and

consistent navigation keep frequent tasks (create order, take payment) within two taps of the Dashboard.

4.2.2 Reliability

Availability $\geq 99.5\%$ during operating hours. MTTR ≤ 30 minutes. The system auto-recovers committed order and payment data after a restart. Order and payment totals calculated with 100% accuracy. No critical defect (loss of committed order/payment data, or inability to take payment) remains in a production release.

4.2.3 Performance

Screen transitions and product searches respond within 1 second on average, 3 seconds at peak. Saving an order or processing a payment completes within 2 seconds on average. Supports ≥ 20 concurrent users and ≥ 60 orders/hour at peak without degradation. Stores ≥ 2 years of order/sales history while maintaining the above response times.

4.2.4 Security

Passwords stored hashed. All transmission of payment and personal data uses ≥ 256 -bit TLS encryption. Role-based authorization per Section 3.1.3. Audit log of login, payment and data-modification events.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-01	An order must contain at least one product before it can be saved or paid.
BR-02	Order total = sum of (product price $\times$ quantity) for all items, minus any applicable discount.
BR-03	All items in a single order must be paid using a single payment method.
BR-04	A product cannot be ordered when marked unavailable or out of stock.
BR-05	Only a Manager or Administrator can create, modify or deactivate products and prices.
BR-06	A discount may not exceed 50% of the order subtotal unless approved by a Manager.
BR-07	Once an order is paid it cannot be edited; corrections via refund or new order.

BR-08	Ingredient stock is reduced automatically when a paid order contains products that use it.
BR-09	Network transmissions involving payment or personal data require at least 256-bit encryption.
BR-10	An account is locked for 15 minutes after 5 consecutive failed login attempts.
BR-11	Loyalty points accrue only on completed payments; redeemed points cannot exceed the customer's balance. [EDIT — confirm loyalty rule]
BR-12	Stock-In quantity must reference an existing Purchase Order line. [EDIT — confirm]

5.2 Common Requirements

ID	Requirement
CR-01	All mandatory fields shall be marked with a visual indicator (*) and validated before data is saved.
CR-02	All date and time values shall use the system date/time and be displayed in the format DD/MM/YYYY HH:mm.
CR-03	The system shall display a success, warning, or error message after every create, update, delete, login, or payment action.
CR-04	The system shall prevent duplicate usernames within the User entity.
CR-05	The system shall automatically record CreatedAt and UpdatedAt timestamps for all transactional records.
CR-06	Only authenticated users may access system functions. Unauthorized users shall be redirected to the Login screen.
CR-07	Users shall only access functions permitted by their assigned role according to the authorization matrix in Section 3.1.3.

CR-08	Data entered into text fields shall be trimmed of leading and trailing spaces before validation.
CR-09	The system shall require confirmation before performing irreversible actions such as deleting records or cancelling orders.
CR-10	All monetary values shall be displayed in Vietnamese Dong (VND) using thousand separators.
CR-11	The system shall maintain an audit log for login, logout, payment, and administrative actions.
CR-12	The system shall automatically log out inactive users after 30 minutes of inactivity.
CR-13	The system shall support searching and filtering lists where applicable (users, customers, products, orders).
CR-14	All screens shall provide a consistent navigation mechanism and a way to return to the previous screen.
CR-15	The system shall validate all user input before sending data to the backend services.

5.3 Application Messages List

#	Code	Type	Context	Content
1	MSG01	In line	No search result	No search results.
2	MSG02	In red, under text box	Required field empty	The {field} field is required.
3	MSG03	Toast	Order created	Order created successfully.
4	MSG04	Toast	Payment completed	Payment completed successfully.
5	MSG05	Toast	Product saved	Product saved successfully.

6	MSG06	Confirmation dialog	Confirm deletion	Are you sure you want to delete this item?
7	MSG07	Toast	Record deleted	Item deleted successfully.
8	MSG08	In red, under text box	Length exceeds max	Exceed max length of {max_length}.
9	MSG09	In line	Wrong username/password	Incorrect user name or password. Please check again.

5.4 Other Requirements...

There are no additional requirements at this time. Any further requirements identified during development will be documented here and reflected in the Record of Changes.